Supplementary Information

Supplementary Note S1. Transverse minimization and slaving

Let v^* be the normalized contrastive eigenvector, let $V_\perp = \{w : w^T v^* = 0\}$, and write $S = xv^* + y$ with $y \in V_\perp$. For fixed ε, define

$$A := \big(I - W(\varepsilon)\big)\big|_{V_\perp}, \qquad q := P_\perp (v^*)^{\circ 3},$$

where $P_\perp$ is the orthogonal projector onto $V_\perp$ and the superscript $\circ 3$ denotes componentwise cubing. Hypothesis (A2) gives $A \succeq \gamma I$.
For the continuous Hopfield energy

$$E(S) = -\frac{1}{2} S^T W(\varepsilon) S + \sum_i F(S_i), \qquad F'(z) = \tanh^{-1}(z),$$

the Hessian is $H(S) = -W(\varepsilon) + D(S)$, with $D(S) = \mathrm{diag}((1 - S_i^2)^{-1}) \succeq I$. Therefore, for every $w \in V_\perp$ inside the activation cube,

$$w^T H(S) w \geq \gamma \|w\|^2.$$

The energy is thus uniformly strongly convex on each transverse slice, so that the slice has at most one minimizer. At $(x, y) = (0, 0)$ the transverse stationarity equation has derivative A, and the implicit function theorem supplies a unique smooth branch $y^*(x)$ locally. Because $E(-S) = E(S)$, uniqueness implies $y^*(-x) = -y^*(x)$.
Projecting the stationarity equation onto $V_\perp$ and using $W(\varepsilon)v^* = \lambda_W(\varepsilon)v^*$ gives

$$0 = P_\perp \nabla E(xv^* + y) = Ay + \frac{x^3}{3} q + O(x^2\|y\| + \|y\|^3 + x^5).$$

Consequently,

$$\boxed{y^*(x) = -\frac{x^3}{3} A^{-1} q + O(x^5)}, \qquad \|y^*(x)\| \leq \frac{|x|^3 \sqrt{h - g^2}}{3\gamma} + O(x^5),$$

where $g = \sum_i (v_i^*)^4$, $h = \sum_i (v_i^*)^6$, and $\|q\|^2 = h - g^2$. This proves the slaving statement used in the main text. It also shows explicitly why transverse minimization first affects the reduced energy at order x^6.

Supplementary Note S2. Sextic coefficient and normal-form barrier

On the line $S = xv^*$, the saturation expansion is

$$\sum_i F(xv_i^*) = \frac{x^2}{2} + \frac{g}{12} x^4 + \frac{h}{30} x^6 + O(x^8).$$

Combining this with the bilinear term and $\mu(\varepsilon) = \lambda_W(\varepsilon) - 1$ gives

$$E(xv^*) = -\frac{\mu}{2} x^2 + \frac{g}{12} x^4 + \frac{h}{30} x^6 + O(x^8).$$

For $y = O(x^3)$, the terms needed through order x^6 are

$$E(xv^* + y) = E(xv^*) + \frac{x^3}{3} y^T q + \frac{1}{2} y^T A y + O(x^8).$$

Minimization in y inserts $y = -(x^3/3)A^{-1}q + O(x^5)$ and yields

$$U(x; \varepsilon) = -\frac{\mu}{2} x^2 + \frac{g}{12} x^4 + c_6 x^6 + O(x^8), \qquad \boxed{c_6 = \frac{h}{30} - \frac{1}{18} q^T A^{-1} q}.$$

Since $0 \preceq A^{-1} \preceq \gamma^{-1} I$ on $V_\perp$,

$$\frac{h}{30} - \frac{h - g^2}{18\gamma} \leq c_6 \leq \frac{h}{30}.$$

The Level-B model used for the dynamical analysis discards the sextic and higher terms:

$$U_{\mathrm{nf}}(x;\varepsilon) = \frac{g}{12}x^4 - \frac{\mu}{2}x^2.$$

Its wells and barrier are

$$x_{\pm} = \pm\sqrt{\frac{3\mu}{g}}, \qquad \Delta E_{\mathrm{nf}} = \frac{3\mu^2}{4g}.$$

Treating c_6x^6 perturbatively gives the leading relative correction

$$\frac{\Delta E - \Delta E_{\mathrm{nf}}}{\Delta E_{\mathrm{nf}}} = -\frac{36c_6}{g^2}\mu + O(\mu^2).$$

These statements concern the near-bifurcation asymptotic energy reduction. The one-dimensional stochastic dynamics used in the production sweep additionally assume adiabatic transverse slaving and do not quantify noise-induced renormalization from transverse fluctuations.

Supplementary verification note

The supplied deterministic symbolic and numerical audit reproduced all eight reported sanity checks without differences from its archived output. The checks covered the saturation series, normal-form stationary points and barrier, canonical rescaling, the actions of the Hebbian perturbation on v^* and u, the x^6 slaving coefficient, transverse convexity, and a counterexample in which (A2) fails. These are consistency checks, not formal verification; no proof assistant was used.

Supplementary Table S1. Complete production sweep

r	Seed	n	Events	κ_{sim}	CI low	CI high	κ_{exact}	κ_K	Err. exact	Err. K	$\log(S/E)$	$\log(S/K)$	$\log(E/K)$	Est. diff. (%)	t^*
0.1500	77000	6000	6000	0.51535	0.44072	0.60261	0.44985	0.19373	0.1456	1.6602	0.1359	0.9784	0.8425	22.856	8.00
0.1938	87000	6000	6000	0.39050	0.38043	0.40084	0.35971	0.18542	0.0856	1.1060	0.0821	0.7448	0.6626	12.990	0.50
0.2504	97000	6000	6000	0.31389	0.30541	0.32261	0.28629	0.17522	0.0964	0.7914	0.0920	0.5830	0.4910	15.080	1.00
0.3235	107000	6000	6000	0.23487	0.22894	0.24094	0.22653	0.16287	0.0368	0.4421	0.0361	0.3661	0.3299	9.847	0.50
0.4180	117000	6000	6000	0.18619	0.18153	0.19098	0.17788	0.14819	0.0468	0.2565	0.0457	0.2283	0.1826	8.615	0.50
0.5400	127000	6000	6000	0.14068	0.13716	0.14429	0.13823	0.13116	0.0177	0.0726	0.0176	0.0701	0.0525	6.832	0.50
0.6977	137000	6000	6000	0.11418	0.11130	0.11713	0.10586	0.11202	0.0786	0.0192	0.0756	0.0191	−0.0566	9.850	1.00
0.9015	147000	6000	6000	0.08287	0.08079	0.08500	0.07936	0.09137	0.0442	0.0931	0.0433	−0.0977	−0.1410	7.696	1.00
1.1647	157000	6000	5999	0.06265	0.06106	0.06427	0.05764	0.07023	0.0869	0.1079	0.0833	−0.1142	−0.1975	9.838	2.00
1.5049	167000	6000	5992	0.04246	0.04139	0.04356	0.03991	0.04998	0.0640	0.1504	0.0620	−0.1630	−0.2250	7.234	2.00
1.9443	177000	6000	5903	0.02664	0.02597	0.02733	0.02573	0.03221	0.0357	0.1727	0.0350	−0.1896	−0.2246	2.691	1.00
2.5121	187000	6000	5400	0.01506	0.01466	0.01547	0.01492	0.01825	0.0094	0.1751	0.0094	−0.1925	−0.2018	1.673	1.00
3.2456	197000	6000	4044	0.00739	0.00717	0.00762	0.00743	0.00877	0.0051	0.1568	−0.0052	−0.1706	−0.1654	0.000	0.00
4.1934	207000	6000	2144	0.00293	0.00281	0.00306	0.00300	0.00340	0.0222	0.1378	−0.0225	−0.1483	−0.1258	0.000	0.00
5.4179	217000	6000	788	0.00094	0.00087	0.00100	0.00091	0.00100	0.0263	0.0633	0.0259	−0.0654	−0.0914	0.000	0.00
7.0000	227000	6000	171	0.000193	0.000166	0.000225	0.000192	0.000205	0.0061	0.0578	0.0061	−0.0596	−0.0656	0.339	0.50

Notes. S, E, and K denote stabilized simulation, numerically evaluated exact one-dimensional MFPT rate, and Kramers asymptotic rate. The relative differences in the Err. columns are absolute. The estimator difference is $100\,|S - S_0|/S_0$, where S_0 is the simulation hazard estimate from $t = 0$ without truncation. Confidence intervals use the post-truncation event count selected by the same hazard-stabilization procedure as the central estimate. Values are rounded for display; machine-precision values are available in `data/final_sweep_data_CORRECTED_CI.csv`.